

Table 3: Effects of AI adoption on employment

	(1)	(2)	(3)	(4)
	Empl. (log)	Empl. (log)	Empl. (log)	Empl. (log)
	No controls	With controls	No controls	With controls
AI adoption (actual)	0.125*** (0.006)	0.079*** (0.006)		
AI adoption (US-based)			0.053*** (0.010)	-0.012 (0.010)
Investment (lag)		-0.003 (0.003)		-0.003 (0.003)
Profitability (lag)		0.153*** (0.019)		0.167*** (0.022)
Financial leverage (lag)		0.034** (0.014)		0.045*** (0.015)
Total assets (log, lag)		0.178*** (0.003)		0.189*** (0.003)
Size FE	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes
Country-sector-wave FE	Yes	Yes	Yes	Yes
Observations	55020	36259	45340	30446
R-squared	0.915	0.926	0.912	0.924
Adjusted R-squared	0.912	0.922	0.908	0.920

Note: The dependent variable is the log of number of employees and is derived from EIBIS. AI adoption (actual) is the EIBIS-reported AI implementation status. AI adoption (US-matched) is the AI implementation status derived from similar firms in the US. Investment is expressed as the annual change in total fixed assets. Profitability is the ratio of EBIT to total assets. Financial leverage is the ratio of loans and long-term debt to total assets. All control variables come from ORBIS and are lagged by 1 year. Standard errors clustered at the country-sector-wave level are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.

We next examine whether the improvement in labor productivity stems from a reduction in employment. The results indicate that this is not the case. Table 3 reports the results, following a structure similar to Table 2, but with the logarithm of employment as the dependent variable. When using the actual series of AI adoption, the estimates suggest a positive effect on employment of 8% without controls (column (2)). In con-

trast, when instrumenting AI adoption using our PSM strategy and controlling for firm characteristics, we find no significant effect on employment (column (4)).

Taking the results from Tables 2 and 3 together begs the question whether the increase in labor productivity translates into benefits for workers. Table 4 addresses this issue by replicating our main regression with controls and all fixed effects, but using either the total wage bill or the average wage per employee, both at the firm level. The results show that AI-adopting firms display higher wages (overall and per employee) than non-adopting firms, after controlling for age, size and country-sector-wave fixed effects. The results suggest that productivity gains have so far benefited employees. It remains to be seen whether the patterns will hold in the medium to long term.

Our empirical analysis yields three key insights regarding AI adoption by non-financial firms in Europe. First, the instrumental variable estimates confirm a positive causal effect of AI adoption on labor productivity. AI adoption raises productivity by 4%, an effect that remains statistically significant even after controlling for firm-specific characteristics such as investment intensity, profitability, leverage and size.¹⁰

While direct comparisons with the macroeconomic literature are not straightforward – given that our results focus on labor productivity and reflect a one-off effect, whereas macro studies typically estimate long-run impacts on total factor productivity (TFP) – our findings appear more consistent with mid-range projections (e.g., [Bergeaud \(2024\)](#)’s 0.3 percentage points annual TFP gains or [Aghion and Bunel \(2024\)](#)’s 0.7 percentage points). The magnitude suggests that AI acts as a complementary input that enhances efficiency, though its aggregate impact is moderated by implementation frictions and skill gaps, contrasting with more optimistic projections of transformative productivity booms (e.g., [Baily, Brynjolfsson, and Korinek \(2023\)](#)’s 1.7 percentage points scenario). The increase in the coefficient from 2.8% (before matching) to 4.1% (after matching) indicates that endogenous adoption – whereby not all productive firms self-select into AI – understates perceived benefits. After isolating exogenous variation by benchmarking with the US adoption rates, the net effect remains positive, significant and economically plausible, reflecting incremental gains from task automation and cognitive augmentation.

¹⁰In the baseline specification, labor productivity is expressed as turnover per employee. However, the effect remains significant at 3.4% when use instead value added per employee (see Table B.3 in the Appendix).

Second, we find no evidence that AI adoption reduces employment in the short run, though longer-term effects remain uncertain. While our naïve specifications associate AI with 8% higher employment, the relationship vanishes after using predicted AI adoption. Moreover, our results suggest that on average the employees of AI-adopting firms benefit in terms of higher wages. This pattern alleviates near-term concerns about job destruction but raises questions about wage inequality. Indeed, benefits may accrue disproportionately to skilled workers coordinating AI-augmented workflows.¹¹

The divergence between significant productivity gains and null employment effects suggests that AI primarily enables capital deepening. This implies that firms increase worker output without reducing headcount, consistent with micro-level evidence that AI tools accelerate task completion without displacing labor (e.g., [Noy and Zhang \(2023\)](#)’s 40% time reduction in writing tasks; [Baily, Brynjolfsson, and Korinek \(2023\)](#)’s 14% productivity lift in customer support). To further support the capital deepening mechanism, we also test whether AI-adopting firms are more innovative. Table [B.4](#) in the Appendix presents the results, based on both actual adoption and our instrumented version, including firm-level controls and the same fixed effects as before. AI-adopting firms are more likely to develop products, services, or processes that are new for the firm (columns 1-2), country (columns 3-4) or global frontier (columns 5-6).

Third, in Table [5](#) we break down the effects of AI adoption by firm size groups. This heterogeneity analysis reveals that AI’s productivity benefits are concentrated. Medium and large firms exhibit stronger gains, highlighting the role of scale in absorbing integration costs and accessing complementary assets (e.g., data infrastructure, technical talent). This is also consistent with the important role that midcaps play in Europe ([EIB, 2024](#)). The disparities point to risks of widening productivity gaps between firms and regions. This echoes the warnings in [Cazzaniga et al. \(2024\)](#) about AI exacerbating inequality.

¹¹We also test whether the impact on employment varies by country using various indicators from the OECD employment protection database. We find no heterogeneous effects for the EU. The results are available upon request.